Neural NLP Pipeline for the BBC Urdu Corpus

DS 5007: Natural Language Processing
Assignment 2

Submitted by

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Repository: https://github.com/SarmadRazaq/i24-7709-NLP-Assignment2



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1. Overview

This assignment continues the Assignment 1 pipeline on 300 BBC Urdu articles (about 460k cleaned tokens). Part 1 builds TF-IDF, PPMI and Skip-gram embeddings, Part 2 uses them in BiLSTM POS and NER taggers, and Part 3 trains a Transformer encoder for 5-class topic classification. All code is from scratch in PyTorch/NumPy, with no pretrained models or built-in Transformer modules. Metadata.json has no section labels, so articles were assigned to the five categories by keyword scoring (Politics 48, Sports 94, Economy 12, International 55, Health & Society 91). POS and NER labels are silver annotations from a rule-based tagger built on the Assignment 1 stemmer, using a lexicon (NOUN 288, VERB 330, ADJ 235, ADV 266 entries) and a gazetteer (111 Pakistani persons, 124 locations, 85 organisations). The 12th POS tag is PRT (particles); PUNC never occurs because cleaned.txt has no punctuation.

2. Part 1 Results: Word Embeddings

2.1 TF-IDF and PPMI. The term-document matrix (300 × 10002: top 10,000 tokens plus <UNK> and <PAD>) uses TF = count / length and IDF = log[N / (1 + df)]. Top TF-IDF words are clearly topical (Sports: cricket, match, runs, player; Politics: police, court, assembly, minister). PPMI (window k = 5) neighbours are good for collocates (court → supreme, judge, hearing) but general nouns pick up function words. In the t-SNE plot of the 200 most frequent words, function words, auxiliaries, numbers and sports terms form separate clusters. ppmi_matrix.npy keeps the 5,000 most frequent words (GitHub size limit).

2.2 Skip-gram Word2Vec. Separate centre (V) and context (U) matrices, K = 10 negatives from f(w)^0.75, d = 100, k = 5, Adam 10^{-3}, batch 1024, 15 epochs; the loss falls from about 3.3 to 2.51 and ½(V + U) is saved. Neighbours of the eight query words are topically coherent (Adalat → hearing, court, lawyer; Maeeshat → import, trade, price), and 8/10 analogies are solved in the top 3 (e.g. Pakistan : Lahore :: India : Delhi; man : woman :: boy : girl). The relations captured are real but mostly topical: capital-city and kinship analogies fail, and the stemmer merges Aabadi with the -abad of city names.

Table 1. Four-condition comparison.

Condition	MRR (20 pairs)	Analogies (top 3)
C1 PPMI	0.431	4/10
C2 SG raw.txt	0.237	4/10
C3 SG cleaned.txt	0.229	8/10
C4 SG d = 200	0.263	7/10

2.3 Comparison. PPMI has the highest MRR because most gold pairs are direct collocates, which PPMI measures, but it solves the fewest analogies. C2's MRR is marginally above C3's (0.237 vs 0.229), far less than one rank on 20 pairs, yet its neighbours split across inflected forms and it solves half as many

analogies, so cleaning helps. Doubling d (C4) raises MRR slightly but solves one analogy fewer: 460k tokens cannot fill 200 dimensions, so larger d does not clearly help. C3 is the best overall and initialises Part 2.

3. Part 2 Results: POS Tagging and NER

3.1 Setup. 500 sentences (125 each from Politics, Sports, International, Health & Society) were split 70/15/15 by topic (7757/1838/1658 tokens); NOUN, POST and VERB are 75% of POS tags and there are 115 PER, 203 LOC, 37 ORG, 54 MISC entities. The 2-layer BiLSTM (128 units per direction, $h = [\rightarrow h \,\|\, \leftarrow h]$, dropout 0.5) starts from C3 embeddings and uses Adam (10^{-3}, weight decay 10^{-4}) with early stopping on validation F1 (patience 5). POS uses softmax with cross-entropy; NER uses a CRF with learnable transitions and Viterbi decoding. Padding is masked out of both losses.

Table 2. Frozen vs fine-tuned embeddings (%).

Task	Embeddings	Val F1	Test acc. / precision	Test macro-F1 / entity F1
POS	frozen	65.1	80.5	68.3
POS	fine-tuned	84.2	94.3	92.1
NER	frozen	52.4	50.0	47.2
NER	fine-tuned	69.3	74.3	63.4

3.2 POS. Fine-tuning beats freezing by 16–24 F1 points, since topic-based vectors must be reshaped for syntax. The POS tagger reaches 94.3% accuracy and 92.1 macro-F1. The most confused pairs, NOUN–ADJ (45), NOUN–ADV (14) and NOUN–VERB (8), are all open-class: stemming strips the endings that separate nominal adjectives from nouns, and conjunct verbs (noun + kar/ho) blur NOUN and VERB.

Table 3. NER test results, BiLSTM-CRF (conlleval, %).

Type	Precision	Recall	F1	Gold entities
PER	27.3	21.4	24.0	14
LOC	100.0	76.9	87.0	26
ORG	100.0	50.0	66.7	2
MISC	66.7	40.0	50.0	5
overall	74.3	55.3	63.4	47

3.3 NER. Locations recur and are learned well; person names are mostly unseen and have the noisiest labels. Without the CRF, entity F1 drops from 63.4 to 62.0 (3-seed means) and the seed standard deviation almost triples, as softmax emits invalid tags such as O → I-PER. False negatives are rare names (Beijing for the Chinese government, Brazil as a team, an unseen journalist) and one boundary error (Khan for Imran Khan); three of five false positives are real names the gazetteer missed.

Table 4. Ablations, test mean ± std over 3 seeds (%).

Setting	POS macro-F1	NER entity F1
Full model (fine-tuned)	92.9 ± 0.6	63.4 ± 1.4
A1 unidirectional LSTM	88.9 ± 1.1	53.6 ± 3.9
A2 no dropout	93.1 ± 0.4	64.3 ± 1.7
A3 random embedding init	86.8 ± 0.6	58.1 ± 1.7
A4 softmax instead of CRF	n/a	62.0 ± 3.8

3.4 Ablations. A1 hurts most (POS -4.0, NER -9.8): Urdu is head-final, so postpositions and auxiliaries that identify a word follow it. A2 changes F1 by less than one standard deviation, as early stopping already limits overfitting. A3 costs 6.1 (POS) and 5.3 (NER) points, so Part 1 embeddings transfer.

4. Part 3 Results: Transformer Topic Classifier

4.1 Setup. Articles are cleaned token-ID sequences padded or truncated to 256, split 70/15/15 by category (210/45/45; Economy has 2 test articles). Separate modules implement masked scaled dot-product attention (returning weights), 4-head attention (d_model 128, d_k = d_v = 32, per-head projections, shared output), a ReLU FFN (d_ff 512), a fixed sinusoidal PE buffer and 4 Pre-LN blocks; a learned [CLS] feeds an MLP (128 → 64 → 5). Training: AdamW (5×10^{-4}, wd 0.01), cosine schedule, 50 warm-up steps, 20 epochs.

Table 5. Topic classification on the same 45-article test set (%).

Model	Params	Test acc.	Macro-F1	Best val. epoch	s / epoch
Transformer	2,082,309	64.4	50.8	10	8.75
BiLSTM (C3 init)	1,632,269	66.7	50.4	7	1.89

4.2 BiLSTM vs Transformer. (1) The BiLSTM is 2.2 points more accurate, i.e. one test article, with macro-F1 tied; most errors in the 5 × 5 confusion matrix involve Health & Society and Politics, and no Economy article is correct. (2) The BiLSTM converged sooner (epoch 7 vs 10); the Transformer sits near chance for about five epochs. (3) The BiLSTM is 4.6× faster per epoch: attention builds 257 × 257 maps per head and layer with separate per-head projections, while the LSTM is one fused kernel, linear in length. (4) In the final layer [CLS] attends to topical keywords (cricket, Test, player in a sports article; signed, meeting in an international one); some heads focus on a few tokens, others stay near uniform. (5) For 200–300 articles the BiLSTM is preferable: its inductive bias and pre-trained embeddings suit small data, while the Transformer learns from scratch and overfits (train accuracy about 97%).

5. Conclusion

PPMI is best for direct relatedness, while Skip-gram embeddings transfer better and clearly help tagging when fine-tuned. Backward context and pre-trained initialisation matter most for the BiLSTM, and the CRF adds a small, stable NER gain. The from-scratch Transformer works and is interpretable but, with about 200 training documents, does not beat the BiLSTM. The main limitations are silver labels, a tiny Economy class and stemmer over-merging.